Chapter 9: Gaussian Processes and the Infinite-Width Limit

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

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Abstract

A Gaussian process is a probability distribution not over numbers but over *functions*: any finite set of its values is jointly Gaussian, with covariance set by a kernel that encodes which inputs should have similar outputs. This makes nonparametric regression exact and Bayesian — condition on data and the posterior is again a GP, giving not just a predicted curve but a calibrated uncertainty that is small near the data and large away from it. The chapter closes with the result that ties this back to deep learning: a neural network of infinite width, with random weights, *is* a Gaussian process. The central limit theorem does it — each output is a sum of many independent random contributions — so the Bayesian prior implied by a wide network is a GP with a kernel you can write down.

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1 A distribution over functions

Definition 1.1 (Gaussian process)**.** A **Gaussian process** $f \sim \mathcal{GP}(m, k)$ is a collection of random variables $\{f(x)\}$ such that any finite subset $(f(x_1), \dots, f(x_n))$ is jointly Gaussian, with mean $m(x_i)$ and covariance $\mathrm{Cov}(f(x_i), f(x_j)) = k(x_i, x_j)$.

The **kernel** k is the whole model. The common choice, the **RBF** (squared-exponential) kernel $k(x, x') = \exp(-\|x - x'\|^2/2\ell^2)$, says nearby inputs are strongly correlated and distant ones nearly independent, so samples from the prior are smooth — the left panel draws five of them. A GP is nonparametric: it does not fix a functional form with finitely many parameters, it places a prior directly on the space of functions through the kernel.

2 Regression is conditioning

Because everything is jointly Gaussian, learning is just conditioning a multivariate normal (Part I, Chapter 4). Given training inputs X with noisy outputs y and a test point $x_\star$, stack them; the joint is Gaussian, and the conditional $f(x_\star) \mid y$ is Gaussian with closed-form mean and variance.

Theorem 2.1 (GP predictive distribution)**.** With kernel matrix $K = k(X, X)$, noise σ^2, and cross-covariance $k_\star = k(x_\star, X)$,

$$\mathbb{E}[f(x_\star) \mid y] = k_\star (K + \sigma^2 I)^{-1} y, \qquad \mathrm{Var}[f(x_\star) \mid y] = k(x_\star, x_\star) - k_\star (K + \sigma^2 I)^{-1} k_\star^\top .$$

Proof. $(f(x_\star), y)$ is jointly Gaussian with covariance blocks $k(x_\star, x_\star)$, $k_\star$, and $K + \sigma^2 I$. The Gaussian conditioning formula — for jointly normal (a, b), $\mathbb{E}[a \mid b] = \mu_a + \Sigma_{ab}\Sigma_{bb}^{-1}(b - \mu_b)$ and $\mathrm{Var}[a \mid b] = \Sigma_{aa} - \Sigma_{ab}\Sigma_{bb}^{-1}\Sigma_{ba}$ — applied with $\mu = 0$ gives exactly the two expressions. □

The middle panel shows the result: the posterior mean interpolates the data, and the ±2-standard-deviation band pinches to nearly zero at each observation and balloons where there are none. This calibrated "I don't know away from the data" is what a plain network's point prediction lacks — and it comes for free from conditioning a Gaussian. The cost is the $(K + \sigma^2 I)^{-1}$, an $n \times n$ solve, which is why exact GPs are reserved for smaller n.

3 A wide network is a Gaussian process

The link to deep learning is a theorem of Neal (1996). Take a one-hidden-layer network with H units,

$$f(x) = \frac{1}{\sqrt{H}} \sum_{j=1}^{H} v_j \, \phi(w_j^\top x + b_j),$$

and put independent zero-mean priors on the weights v_j, w_j, b_j.

Theorem 3.1 (Infinite-width networks are GPs)**.** As $H \to \infty$, the function f converges to a Gaussian process. For any inputs, the outputs $(f(x_1), \ldots, f(x_n))$ are jointly Gaussian with covariance $k(x, x') = \mathbb{E}\big[v^2\big]\, \mathbb{E}_{w,b}\big[\phi(w^\top x + b)\phi(w^\top x' + b)\big]$.

Proof sketch. For fixed inputs, $f(x)$ is a sum of H i.i.d. terms $v_j \phi(w_j^\top x + b_j)$, each with mean zero and finite variance, scaled by $1/\sqrt{H}$. By the central limit theorem (Part I, Chapter 5) the sum converges to a Gaussian; applied jointly to several inputs (the multivariate CLT) the whole vector is jointly Gaussian, so f is a GP. Its covariance is the per-unit expectation above, since the units are independent. □

So the central limit theorem of Part I resurfaces as a statement about neural networks: the prior over functions implied by a wide, randomly-initialised network is a Gaussian process with a specific kernel — the **neural network Gaussian process** (NNGP) kernel. The right panel is the theorem happening: the distribution of a single output over random weights is sharply non-Gaussian at width 1, and by width 200 it has become the Gaussian the CLT predicts. This is the bridge between the two halves of the course — the Bayesian, kernel view of a neural network is literally a Gaussian process — and it is why GPs, kernels, and deep networks are studied together.

Summary

A Gaussian process is a distribution over functions: any finite set of values is jointly Gaussian with covariance given by a kernel k, which encodes smoothness (e.g. the RBF kernel). Regression is Gaussian conditioning (Chapter 4): the predictive mean $k_\star(K + \sigma^2 I)^{-1} y$ interpolates the data and the predictive variance gives calibrated uncertainty, small at the data and large away from it, at the cost of an $n \times n$ solve. The tie to deep learning is Neal's theorem: a one-hidden-layer network with random weights and $1/\sqrt{H}$ scaling converges, as $H \to \infty$, to a Gaussian process by the central limit theorem (Chapter 5), with the NNGP kernel as its covariance — the Bayesian prior of a wide network is a GP.